Clinicflow overview and Progressive Building plan

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Consolidated recommendations, risks, architechture reviews, UI/UX improvements, and testing strategies.

ClinicFlow's mission:

A reliable, simple, affordable hospital and clinic management platform built specifically for Nigerian clinics that need software that works even with poor internet and old hardware.

Current state;

1. Admission/discharge history display
2. Search-first patient matching
2. Separate complaint from diagnosis
3. Age → Date of Birth
6. Authentication and role-based access
5. Patient management and visit history
7. Queue and triage system
8. Bed management and admissions workflow
9. Multi-tenancy hardening
10. Admin dashboard and staff management
11. Production deployment on Vercel, Render, and Supabase
12. Clear product vision and business strategy

LONG TERM ECOSYSTEM:

MOI-DOCTAR (AI triage)

↓

ClinicFlow

↓

Doctor & Clinic Operations...

(MOI-DOCTAR becomes the patient-facing entry point, while ClinicFlow becomes the clinic's operating system.).

TECH STACK:

Layer	Technology	Status
Frontend	React + TypeScript + Vite	LIVE — Vercel
Styling	Tailwind CSS v3	v3 pinned (D17). Live.
Routing	React Router v6	Live — Outlet context
Server State	React Query (TanStack)	LIVE — real data
Backend	Node.js + Express + TypeScript	LIVE — Render
Auth	JWT (httpOnly cookies) + Bcrypt	LIVE — cross-site cookie fixed (D25)
Database	PostgreSQL + Prisma 6.x	LIVE — Supabase, Frankfurt
Frontend Deploy	Vercel	LIVE (free tier; auto-deploys)
Backend Deploy	Render	LIVE (free tier; Manual Deploy; D1 pending)
DB Host	Supabase	LIVE (free tier; D1 pending)
Payments	Paystack	Phase 3 (manual invoicing first)
SMS / WhatsApp	Termii / WhatsApp Business API	Phase 4
Biometric Verify	Smile ID / Youverify (API)	Phase 3/4 — never store faces

CLINICFLOW MASTER TESTING PLAN

Mission and Assignment For Testers

Verify that ClinicFlow is:

1. Reliable
2. Secure
3. Fast
4. Multi-tenant safe
5. Resistant to user mistakes
6. Production-ready for real clinics

TESTING PHASE 1 — AUTHENTICATION & ACCOUNT SYSTEM

A) "Registration"

Test Cases;

1. Create clinic with valid data.
2. Register with empty fields.
3. Register with invalid email.
4. Register with weak password.
5. Register with duplicate email.
6. Register with very long clinic name.

7. Register with special characters.

Verify that...

1. Password is hashed.
2. User automatically belongs to correct clinic.
3. JWT cookie created.
4. No duplicate clinics created.

B) "Login"

Test Cases;

1. Correct credentials.
2. Wrong password.
3. Wrong email.
4. Empty fields.
5. Extremely long inputs.
6. SQL injection attempts.

Verify that...

1. No information leakage.
2. Proper error messages.
3. Session created correctly.

C) "Logout"

Test Cases;

1. Logout once.

2. Logout twice.

3. Logout after cookie expiry.

Verify that....

1. Cookie removed.

2. Protected routes inaccessible.

TESTING PHASE 2; MULTI-TENANCY (VERY VERY VERY "IMPORTANT")

If this fails, the product cannot go live.

A) Scenario

Create:

"Clinic A"

Create:

Patients

Beds

Staff

Create:

"Clinic B"

Attempt to access:

1. /api/patients/id
2. /api/beds/id
3./api/staff/id.

from Clinic B..

<u>Expected;</u>

404

403

<u>NEVER:</u>

data returned

partial information

patient names exposed.

<u>B) Direct URL Testing</u>

<u>C) API Testing</u>

TESTING PHASE 3; PATIENT REGISTRATION

A) Create Patient

Test;

1. Valid data

2. Empty data

3. Very long names

4. Nigerian phone numbers

5. Missing phone

6. Missing age/DOB

7. Special characters

B) Duplicate Patients

Verify that....

1. Search appears.

2. Duplicate prevention works.

Aditional.... try admitiing patients with special names

TESTING PHASE 4; CHECK-IN FLOW

Test;

Patient:

Check in

Check in twice

Check in yesterday

Check in after midnight

Verify....

Queue order.

Emergency should always appear first.

TESTING PHASE 5; ADMISSION HISTORY

Verify...

- Admit note saved.
- Discharge note saved.
- Timestamps correct.
- User names correct.
- Bed snapshots preserved.

TESTING PHASE 6; STAFF MANAGEMENT

Create Staff

- Test:
  - Duplicate emails.
  - Invalid roles.
  - Remove staff.

TESTING PHASE 7; ROLE PERMISSIONS

Receptionist Must NOT:

- Admit patient.
- Discharge patient.
- Access admin dashboard.

Doctor Must:

- Admit.
- Discharge.

Mark seen.

Admin Can access everything

TESTING PHASE 8; SECURITY TESTING

TESTING PHASE 9; PERFORMANCE

Queue

Create:

1000 patients.

Measure:

1. Loading time
2. Searching
3. Check-in speed

Beds

Create:

500 beds.

Staff

Create:

200 users.

TESTING PHASE 10; NETWORK FAILURE TESTING

we know the country wey we dey):

Simulate:

Slow internet.

Network loss.

Backend sleep.

Timeout.

Browser refresh.

Expected:

No data loss.

Proper error messages.

Recovery possible.

TESTING PHASE 11; MOBILE TESTING

Devices:

Small Android

Tablet

Old laptop

Verify:

Bottom nav.

Forms.

Tables.

Modals.

TESTING PHASE 11; DATA INTEGRITY

1

Create complete patient journey:

Register

↓

Check-in

↓

Called

↓

Seen

↓

Admitted

↓

Discharged

↓

Return next month

↓

Check-in again

Verify:

1. Same patient ID.
2. History preserved.

3. No duplicate records.
4. Previous admissions visible.

RELEASE CRITERIA

ClinicFlow is ready for Clinic #1 only if:

✅ No cross-clinic data leakage.

✅ No data loss.

✅ No duplicate admissions.

✅ No broken patient history.

✅ No unauthorized actions.

✅ System survives bad internet.

✅ All critical workflows pass.

In case of BUGS, document them in grades as...

P0 – Patient safety/data leak → fix immediately.

P1 – Core workflow broken.

P2 – Major UX issue.

P3 – Cosmetic issue.

UI/UX IMPROVEMENTS….

1)Reception Dashboard

There should be a change, so it now looks like this

```
┌─────────────────────────────┐
│ Add Patient                 │
├─────────────────────────────┤
│ Waiting Patients            │
├─────────────────────────────┤
│ Today's Statistics          │
└─────────────────────────────┘
```

Current issue:
Receptionists have to mentally switch contexts.
The dashboard should feel like an air traffic control screen.

2)Universal Search Bar

Top bar:
⍰ Search patients, visits, admissions...

Should instantly search:
patients
phone numbers
patient IDs
admissions.....

3)Patient Quick Card

Instead of:
Andrew Stephen Enobong

__Show:__
Andrew Stephen Enobong
Male • 31 yrs
07014******
O-
AA
⚠ Hypertension (I no get ooo)
⚠ Penicillin Allergy (na just case study lol)::)

{Critical informations should never require another click.}

4)Sticky Safety Banner

When opening patient records:

⚠ PENICILLIN ALLERGY
⚠ DIABETES

Pinned at the top.
Never scroll away.

5)Queue Card Redesign
Instead of:
Andrew Stephen
Urgent

Card:
Emergency
Andrew Stephen
41 • Male
Chest Pain

Waiting: 25 min
Time waiting is incredibly important.

6)Queue Aging Indicators
"Who has been waiting the longest?"
So it looks like;
0-15 mins Green
15-30 mins Yellow
30-60 mins Orange
60+ mins Red

SOME OBSERVATION AND REVIEWS FROM USERS I GATHERED;

1. Patient's profile (phone number, address, guardian, doctor'z report, prescription, referred patient etc
2. Assigning doctors to patients
3. Staff sign in attendance
4. Also instead of making it just... Doctor, receptionist n security, you know there are other departments ry (like sanitation , maintenance, utility, IT, accountant, customer service, room service, researchers etc) Can we just create a different space on the staff profile for that? So they'll fill in their department... Also can we have it all separated after each profile is being filled? Separated by departments (the profile can also take pictures, credentials etc)

5. Can there also be kinda like an SOS that will call the attention of all staff or all departments at once (especially in case of emergency)
6. I don't know if it's fine for the patients to also have access to their profile (now instead of seeing the doctor anyhow,

they'll just check online and can also send signals there)... They'll see their health status, prescriptions, doctor in charge with picture (so no one can impersonate)... The access they'll be having is not into the admins own report own... Just prolly like a page or 2 where all their details are... The can also make payments there ry?

7. Can we also have the following space... Speed contacts for, fire service, waste management, supply of hospital resources etc
8. Can we also have an agreement form? Like an undertaking form? Incase of stuffs yhhhhh?
9. There’s no customer care center on the app
10. A platform for conversation between any patient and the doctor should be made, N.B; the doctor can be in AI form or a real life person speaking. (for the AI doctor, MOI DOCTAR will cover up)

Architecture I'd Recommend Going Forward

<u>Frontend</u>

- React
- React Query
- Context

<u>Backend</u>

- Modular Monolith
  - Auth
  - Patients
  - Queue
  - Beds
  - Admissions
  - Staff
  - Billing
  - Notifications

Database

PostgreSQL

Indexes

Soft Deletes

Audit Logs

Infrastructure

Vercel

Render

Supabase

Background Jobs

Redis

BullMQ

Future

Analytics

SMS

MOI-DOCTAR Integration